

Safety Data Sheet

Clethodim 2E

Date: 02/04/2026

Version No: 1

TENKŌZ

1. IDENTIFICATION

PRODUCT NAME: Clethodim 2E
DESCRIPTION: A liquid herbicide.
EPA REG. NO.: 103591-11-55467
COMPANY IDENTIFICATION: Tenkoz, Inc.
1725 Windward Concourse, Suite
410 Alpharetta, GA 30005
EMERGENCY PHONE: 800-424-9300

2. HAZARD IDENTIFICATION

WARNING

May cause an allergic skin reaction (H317)
Causes skin and eye irritation (H315+H320)
Suspected of causing cancer (H351)
Harmful if swallowed or if inhaled (H302+H332)
Combustible liquid (H227)
Harmful to aquatic life (H402)



HAZARD CLASSIFICATION

Health Hazards	Category	Physical Hazards	Category
Sensitization-Skin	1	Flammable Liquids	4
Eye Damage / Irritation	2B		
Skin Irritation	2	Environmental Hazards	Category
Carcinogenicity	2	Hazardous to the Aquatic Environment -- Acute	3
Acute Oral Toxicity	4		
Acute Inhalation Toxicity	4		

HAZARDS NOT REQUIRING CLASSIFICATION

None

PRECAUTIONARY STATEMENTS

Do not handle until all safety precautions have been read and understood. Wear protective gloves, protective clothing, and eye protection. (P201+P202+P280)
Avoid breathing mist/vapors/spray. Contaminated work clothing should not be allowed out of the workplace. Wear protective gloves. (P261+P272+P280)
Do not eat, drink or smoke when using this product. (P270)
Wash hands and exposed skin thoroughly after handling. (P264)
Use only outdoors or in a well-ventilated area. (P271)
Keep away from heat, hot surface, sparks, open flames and other ignition sources. No smoking. (P210)
Avoid release to the environment not in accordance with the product label. (P273)

IF ON SKIN: Wash with plenty of water and soap. If skin irritation or rash occurs: Get medical advice or attention. Refer to Section 4 for additional medical information. Specific treatment. (See Section 4 for additional medical information). Take off contaminated clothing and wash it before reuse. (P302+P352+P333+P313+P321+P362+P364)

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice or attention. Refer to Section 4 for additional medical information. (P305+P351+ P338+P337+P313)

IF exposed or concerned: Get medical advice / attention (P308+P313)

IF SWALLOWED: Call a Poison Control Center or doctor if you feel unwell. Rinse mouth. (P301+P312+P330)

IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a Poison Control Center or doctor if you feel unwell. (P304+P340+P312)

Store in a well-ventilated place. Store locked up. (P403+P405)

Dispose of contents / containers in accordance with local, State and Federal regulations. Refer to the product label for specific disposal instructions. (P501)

3. COMPOSITION / INFORMATION ON INGREDIENTS

Common Name	Chemical Name	CAS #	Composition
Clethodim	(E)-2-[1-[[[(3-chloro-2-propenyl)oxyl]imino]propyl]-5-[2-(ethylthio)propyl]-3-hydroxy-2-cyclohexen-1-one]	99129-21-2	26.4%
Aromatic hydrocarbon	Solvent Naphtha (Petroleum), Heavy Aromatic	64794-5	58-62%
Naphthalene	--	91-20-3	< 6%

NOTE: Ingredients not precisely identified are proprietary or non-hazardous. Values are not product specifications.

4. FIRST AID MEASURES

Have the product container or label with you when calling a poison control center or doctor, or going for treatment. You may also contact the Poison Control Center at **1-800-222-1222** for emergency medical treatment information. For chemical emergency: spill, leak, fire, exposure or accident, call CHEMTREC: **1-800-424-9300**.

IF IN EYES: Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. If eye irritation persists: Get medical attention/advice.

IF ON SKIN OR CLOTHING: Take off contaminated clothing. Rinse skin immediately with water for 15-20 minutes. Call a doctor or poison control center if you feel unwell.

IF SWALLOWED: Immediately call a poison control center or doctor. Do not induce vomiting unless told to do so by a poison control center or doctor. Do not give any liquid to the person. Do not give anything by mouth to an unconscious person.

IF INHALED: Move person to fresh air. If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice.

NOTE TO PHYSICIAN: Ingestion of this product or subsequent vomiting can result in aspiration of light hydrocarbon liquid, which can cause pneumonitis. Contains petroleum distillate.

5. FIREFIGHTING MEASURES

Flash Point (Closed Cup): 142°F (61.1°C)

Flammable Limits: Unknown

Fire and Explosion Hazards: Combustible liquid. Can form explosive mixtures at temperatures at or above the flashpoint. In fire can burn and form explosive vapors from ruptured containers.

Extinguishing Medium: Water fog, foam, dry chemical or carbon dioxide (CO₂). Do not use water jet as this will spread the fire.

Fire Fighting Equipment: Firefighters should be equipped with self-contained positive pressure breathing apparatus and full bunker gear.

Fire Fighting Instructions: Evacuate nonessential personnel to prevent exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated buildings, area and equipment until decontaminated. Dike and collect any runoff to prevent entry to drains or water bodies.

Hazardous Combustion Products: Thermal decomposition will produce nitrogen oxides, sulfur oxides and may produce toxic chlorine compounds.

NFPA Ratings: Health – 1 / Flammability – 2 / Reactivity - 0

6. ACCIDENTAL RELEASE MEASURES

Clean up spills immediately, observing precautions in Section 8 of this SDS. Isolate hazard area and keep unnecessary and / or unprotected personnel from entering the spill area.

Small Spills: Absorb small spills with sand, vermiculite or other inert absorbent material. Place contaminated material in a container for disposal.

Large Spills: Control spill at the source and prevent material from entering soil, sewers, waterways or low areas. Dike large spills using absorbent or impervious material such as clay or sand. Recover materials for reuse or disposal. After removal, clean contaminated area thoroughly with water being sure to recover the wash liquid with additional absorbent material and place in a container for disposal.

7. HANDLING AND STORAGE

Keep out of reach of children.

Handling: Avoid contact with skin. Do not get in eyes, on skin, or on clothing. Harmful if swallowed or inhaled. Avoid breathing vapors or spray mist. Wash thoroughly with soap and water after handling and before eating, drinking, chewing gum using tobacco, or using the toilet.

Storage: Keep pesticide in original container. Do not put concentrate or dilute into food or drink containers. Store in cool, dry place. Do not store diluted spray.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Engineering Controls: Handle only with adequate ventilation. Facilities storing or utilizing this material should be equipped with an eye-wash station and a safety shower.

Protective Clothing: Wear long-sleeved shirt and long pants, shoes, socks, chemical resistant gloves and protective eyewear.

General: Remove PPE immediately after handling this product. Wash the outside of gloves before removing. As soon as possible, wash thoroughly and change into clean clothing. Discard clothing and other absorbent materials that have been heavily contaminated with this product. Do not reuse them. Follow manufacturer's instructions for cleaning and maintaining PPE. If no such instructions for washables, use detergent and hot water. Keep and wash PPE separately from other laundry.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance:	dark amber liquid
Odor:	mild aromatic
Melting/freezing point:	not available
Boiling point/Boiling range:	not available
Flammability:	ignitable
Flammability limits (upper/lower):	not available
Flash point:	142°F (61.1°C)
Auto-ignition temperature:	not available
Decomposition temperature:	not available
pH:	4.0-5.0 (1% solution)
Kinematic viscosity:	not available
Solubility:	emulsifies
Partition coefficient:	not available
Vapor pressure:	not available
Density:	7.78 - 8.26 lb/gal
Relative vapor density:	not available
Particle characteristics:	(solids only)

10. STABILITY AND REACTIVITY

CONDITIONS TO AVOID: Heat, flame and extreme temperatures

CHEMICAL STABILITY: Stable under all normal use and storage conditions. Unstable at extreme pH's, temperatures, and exposure to UV light.

INCOMPATIBILITY WITH OTHER MATERIALS: None known.

HAZARDOUS DECOMPOSITION PRODUCTS: May release nitrogen oxide, sulfur oxides, carbon oxides (CO and CO₂) and aniline when thermally decomposed.

HAZARDOUS POLYMERIZATION: Will not occur.

11. TOXICOLOGICAL INFORMATION

ORAL TOXICITY (rat LD₅₀): > 2,000 mg/kg

DERMAL TOXICITY (rat LD₅₀): > 5,000 mg/kg

INHALATION TOXICITY (rat LC₅₀): > 2.0 mg/L (4-hour)

EYE IRRITATION: Rabbit – Moderately irritating

SKIN IRRITATION: Rabbit – Moderately irritating

SKIN SENSITIZATION: Guinea Pig – Skin sensitizer

CARCINOGENICITY:

EPA: Group C: Possible human carcinogen (naphthalene)

NTP: Reasonably anticipated to be a human carcinogen (naphthalene)

ACGIH: Not Listed

IARC: 2B: Possibly carcinogenic to humans (naphthalene)

OSHA: Not Listed

MUTAGENIC TOXICITY: No evidence of mutagenic effects during *in vivo* and *in vitro* assays.

REPRODUCTIVE TOXICITY: Not known to cause reproductive or birth defects at normal exposure levels.

MEDICAL CONDITIONS AGGRAVATED BY EXPOSURE: Exposure to vapors may increase tendency to respiratory or bronchial infections.

12. ECOLOGICAL INFORMATION

The following information is for the active ingredient, Clethodim:

AQUATIC TOXICITY

Bluegill Sunfish (96hr LC₅₀): >100 mg/L

Rainbow Trout (96hr LC₅₀): 56 mg/L

Daphnia magna (96hr LC₅₀): >100 mg/L

AVIAN TOXICITY

Bobwhite Quail (LD₅₀): > 6,000 ppm

Mallard (LD₅₀): > 6,000 ppm

OTHER

Bees (LD₅₀): > 100 µg/bee

13. DISPOSAL CONSIDERATIONS

PESTICIDE DISPOSAL: Wastes resulting from the use of this product may be disposed of on site or at an approved waste disposal facility.

CONTAINER DISPOSAL: Nonrefillable container. Do not reuse or refill this container. Refer to the product label for specific container disposal instructions.

14. TRANSPORT INFORMATION

US-DOT:

Containers ≤ 119 gal. (450 L):

Not regulated

Containers > 119 gal. (450 L):

Shipping Description:

NA1993, Combustible liquid, N.O.S. (naphthalene), PG III

Hazardous Substance RQ:

100 lbs. (180 gallons of product)

Emergency Guide:

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IMDG:

Not regulated

IATA:

Not regulated

This information is not intended to convey all specific regulatory or operational requirements/information relating to this product. Transportation classifications may vary by container volume and may be influenced by regional or country variations in regulations. Additional transportation system information can be obtained through an authorized sales or customer service representative. It is the responsibility of the transporting organization to follow all applicable laws, regulations and rules relating to the transportation of the material.

15. REGULATORY INFORMATION

FIFRA –

This chemical is a pesticide product registered by the Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets, and for workplace labels of non-pesticide chemicals. The following is the hazard information as required on the pesticide label:

PRECAUTIONARY STATEMENTS

HAZARDS TO HUMANS AND DOMESTIC ANIMALS

WARNING. Causes substantial but temporary eye irritation. Avoid contact with skin. Do not get in eyes, on skin, or on clothing. Harmful if swallowed or inhaled. Avoid breathing vapors or spray mist. Prolonged or frequently repeated skin contact may cause allergic reactions in some individuals.

See inside label booklet for additional Precautionary Statements and Directions for Use including Storage and Disposal instructions.

ENVIRONMENTAL HAZARDS

Do not apply directly to water, to areas where surface water is present, or to intertidal areas below the mean high water mark. Do not apply where runoff is likely to occur. Do not apply where weather conditions favor drift from areas treated. Do not contaminate water when disposing of equipment washwater or rinsate.

NON-TARGET ORGANISM ADVISORY STATEMENT

This product is toxic to plants and may adversely impact the forage and habitat of non-target organisms, including pollinators, in areas adjacent to the treated area. Protect the forage and habitat of non-target organisms by following label directions intended to minimize spray drift.

The use of this product may pose a hazard to the federally designated endangered species of Solano Grass and Wild Rice.

PHYSICAL OR CHEMICAL HAZARDS

Combustible. Do not use or store near heat or open flame.

All pesticides are governed under the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA). The regulatory information presented below is pertinent only when this product is handled outside of the normal use and application as a pesticide. This product is excluded from listing requirements under EPA/TSCA.

SARA Title III – Section 302 Extremely Hazardous Substances

Not listed

SARA Title III – Section 311/312 Hazard Categories

Immediate, Delayed, Fire

SARA Title III – Section 312 Threshold Planning Quantity

N/A

SARA Title III – Section 313 Reportable Ingredients

Naphthalene (CAS No. 91-20-3): < 6%

CERCLA Reportable Quantity (RQ) –

Naphthalene (CAS No.91-20-3): 100 lbs. (180 gallons of product)

CALIFORNIA PROP 65 STATUS –



WARNING: This product can expose you to chemicals including naphthalene, which is known to the State of California to cause cancer. For more information, go to www.P65Warnings.ca.gov.

CANADA –

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the SDS contains all of the information required by CPR.

16. OTHER INFORMATION

This Safety Data Sheet (SDS) serves different purposes than and DOES NOT REPLACE OR MODIFY THE EPA APPROVED PROD-UCT LABELING (attached to and accompanying the product contain-er). This SDS provides important health, safety, and environmental in-formation for employers, employees, emergency responders and others handling large quantities of the product in activities generally other than product use, while the labeling provides that information specifically for product use in the ordinary course.

To the extent consistent with applicable law, neither Tenkoz, Inc. nor Seller be liable for any incidental, consequential or special damages resulting from the use or handling of this product. TO THE EXTENT CONSISTENT WITH APPLICABLE LAW, THE EXCLUSIVE REM-EDY OF THE USER OR BUYER, AND THE EXCLUSIVE LIABILITY OF TENKOZ, INC. AND SELLER FOR ANY AND ALL CLAIMS, LOSSES, INJURIES OR DAMAGES (INCLUDING CLAIMS BASED ON BREACH OF WARRANTY, CONTRACT, NEGLIGENCE, TORT, STRICT LIABILITY OR OTHERWISE) RESULTING FROM THE USE OR HANDLING OF THIS PRODUCT, SHALL BE THE RETURN OF THE PURCHASE PRICE OF THE PRODUCT OR, AT THE ELEC-TION OF TENKOZ, INC. OR SELLER, THE REPLACEMENT OF THE PRODUCT.

SDS Version: 1

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Revisions: New product and company name